Sequences and Series

Pure Mathematics Year 2

Chapter 3

Sequences and Series

Arithmetic sequences and series · geometric sequences and series · sigma notation · recurrence relations

Chapter Overview

1. Arithmetic series

Determine the value of

$2 + 4 + 6 + \cdots + 100$.

2. Geometric series

The first term of a geometric sequence is 3 and the second term is 1. Find its sum to infinity.

3. Sigma notation

Determine the value of

$$\sum_{r=1}^{100} (3r + 1).$$

4. Recurrence relations

If $a_1 = k$ and $a_{n+1} = 2a_n - 1$, determine a_3 in terms of k.

Types of sequences 3

1. $2, 5, 8, 11, 14, \ldots$

Name this type of sequence and describe the change between consecutive terms.

1

2. $3, 6, 12, 24, 48, \ldots$

Name this type of sequence and describe the change between consecutive terms.

2

3. $1, 1, 2, 3, 5, 8, \ldots$

How is each new term generated?

3

Definition

An arithmetic sequence has a constant difference between consecutive terms.

Geometric sequences and recurrence relations will be explored later in this chapter.

The fundamentals of sequences 4

Sequence notation

Consider

$2, 5, 8, 11, 14, \ldots$

What does n represent?

1

What does u_n represent?

2

For the value 8, state its position and its term notation.

3

The general term

Let a be the first term and d the common difference.

Moving to position n takes $n-1$ steps.

$u_1 = a,$

$u_2 = a + d,$

$u_3 = a + 2d.$

$$\boxed{u_n = a + (n-1)d}$$

Write down a, d and the relevant position before substituting.

Example 1

$u_n = 55 - 2n.$

(a) Find the first three terms.

(b) Find the first negative term.

1a

1b

Example 2

Find the general term of each sequence.

(a) $6, 20, 34, 48, 62, \ldots$

(b) $101, 94, 87, 80, 73, \ldots$

2a

2b

Further Examples 6

1. Find the constants

A sequence is generated by $u_n = an + b$, where a and b are constants. Given $u_3 = 5$ and $u_8 = 20$, find a and b.

1

2. Equal differences

For which values of x do -8, x^2 and $17x$ form the first three terms of an arithmetic sequence?

2

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Xin has a 14-day training schedule. She runs for A minutes on day 1 and increases her running time by $d + 1$ minutes each day, where A and d are constants.

(a) Show that on day 14 she runs for $A + 13d + 13$ minutes. (2 marks)

a

Yi also has a 14-day schedule. She runs for $A - 13$ minutes on day 1 and increases her running time by $2d - 1$ minutes each day.

(b) Yi and Xin run for the same time on day 14. Find d. (3 marks)

b

4. The first term of an arithmetic sequence is 14. The fourth term is 32. Find the common difference.

4

5. A sequence is generated by $u_n = pn + q$. Given $u_6 = 9$ and $u_9 = 11$, find p and q.

5

6. An arithmetic sequence has $u_3 = 30$ and $u_9 = 9$. Find its first negative term.

6

7. The 20th term is 14 and the 40th term is -6. Find the 10th term.

7

8. The first three terms are $5p$, 20 and $3p$. Find the 20th term.

8

10. An arithmetic sequence has first term k^2 and common difference k, where $k > 0$. The fifth term is 41. Find k, giving your answer in the form $p + q\sqrt{5}$, where p and q are integers. (4 marks)

10